

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

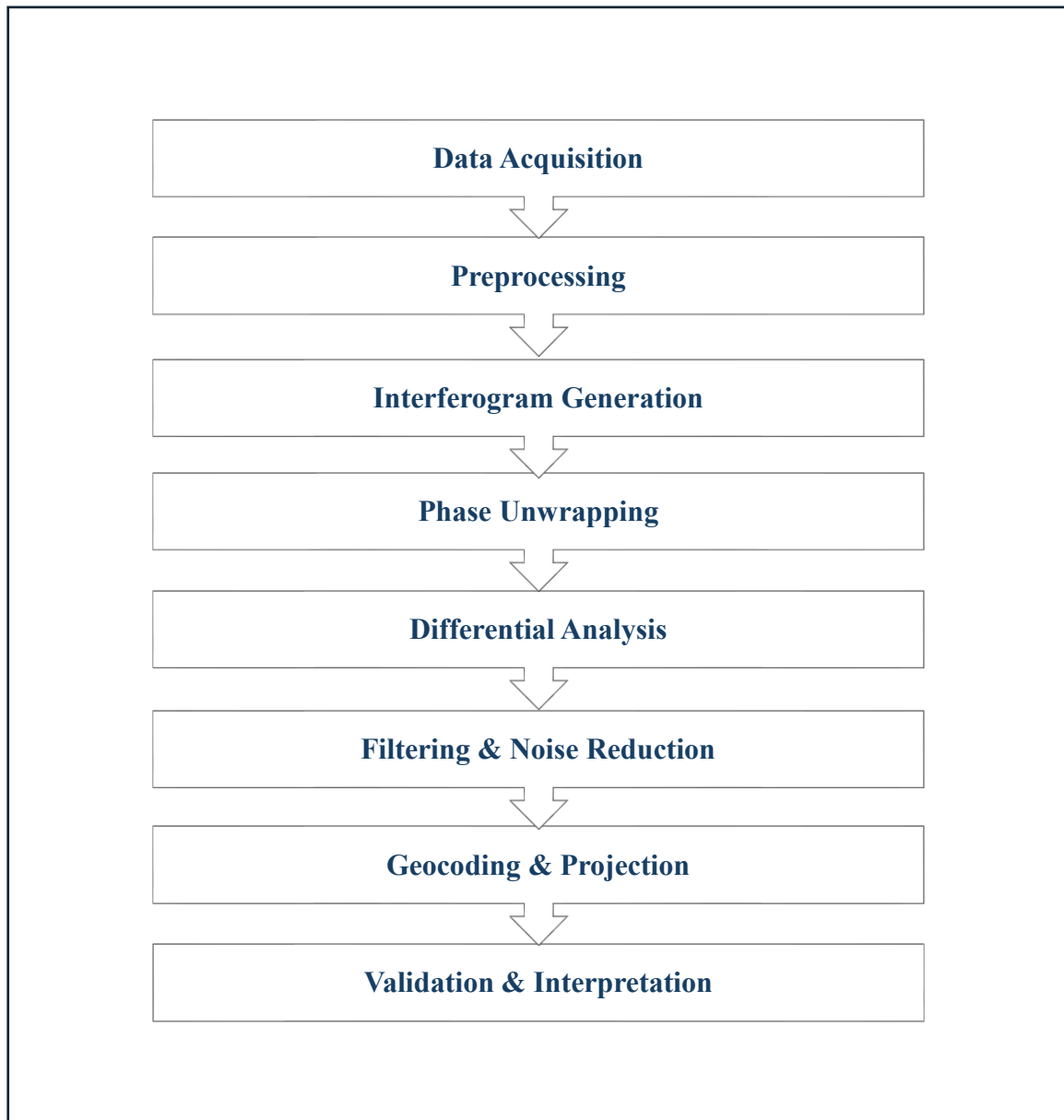
Multi-temporal Sentinel-1 SLC IW mode data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
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S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 926+250

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

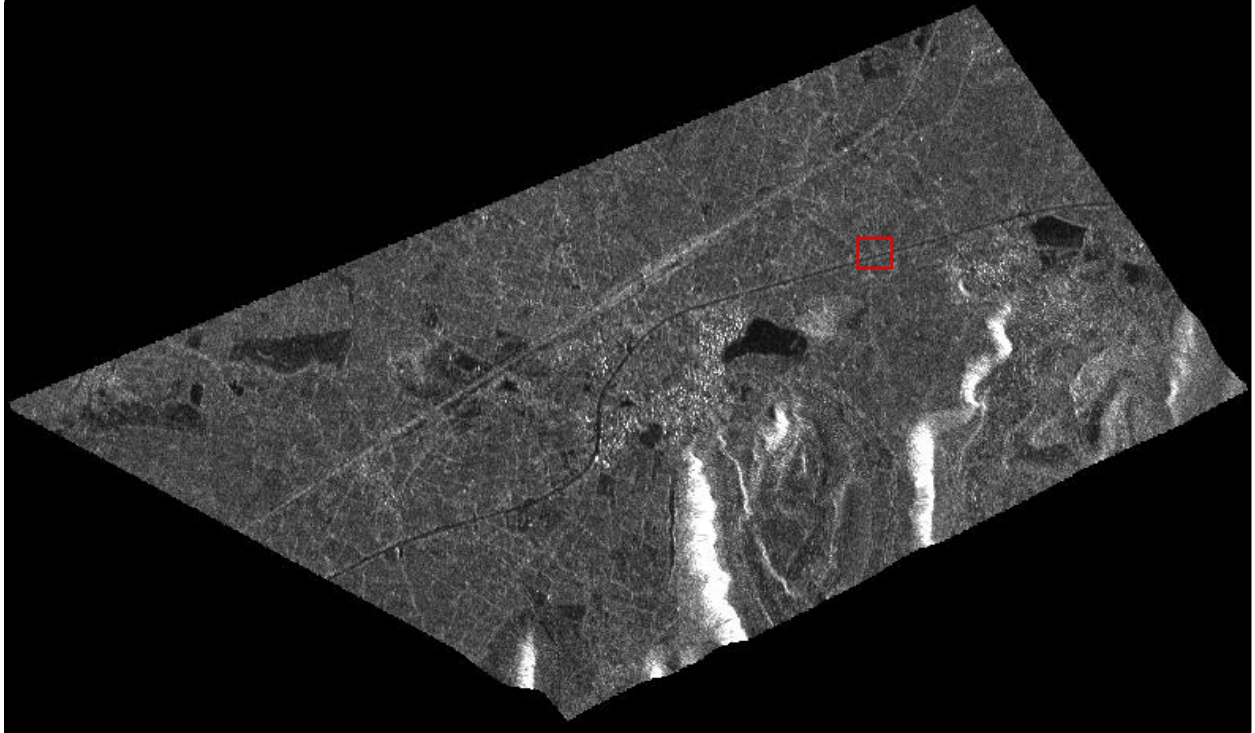
Flight Direction - DESCENDING

Polarization – VV+VH

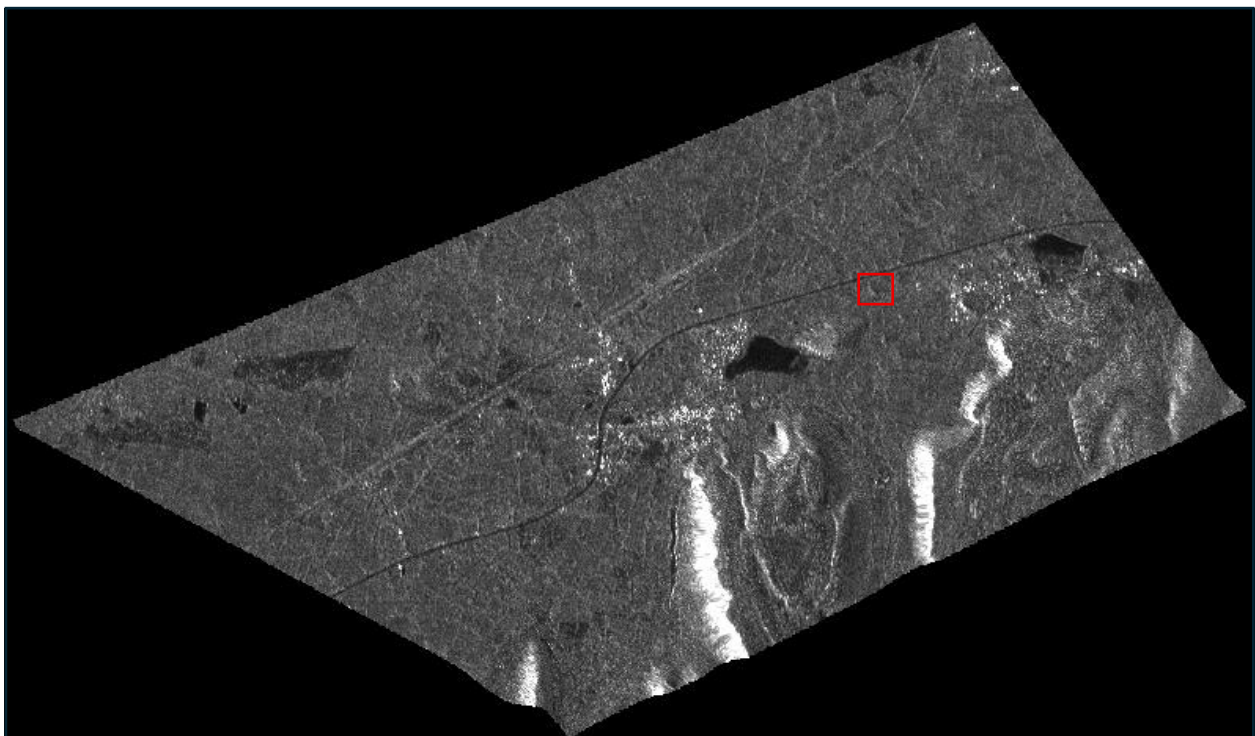
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

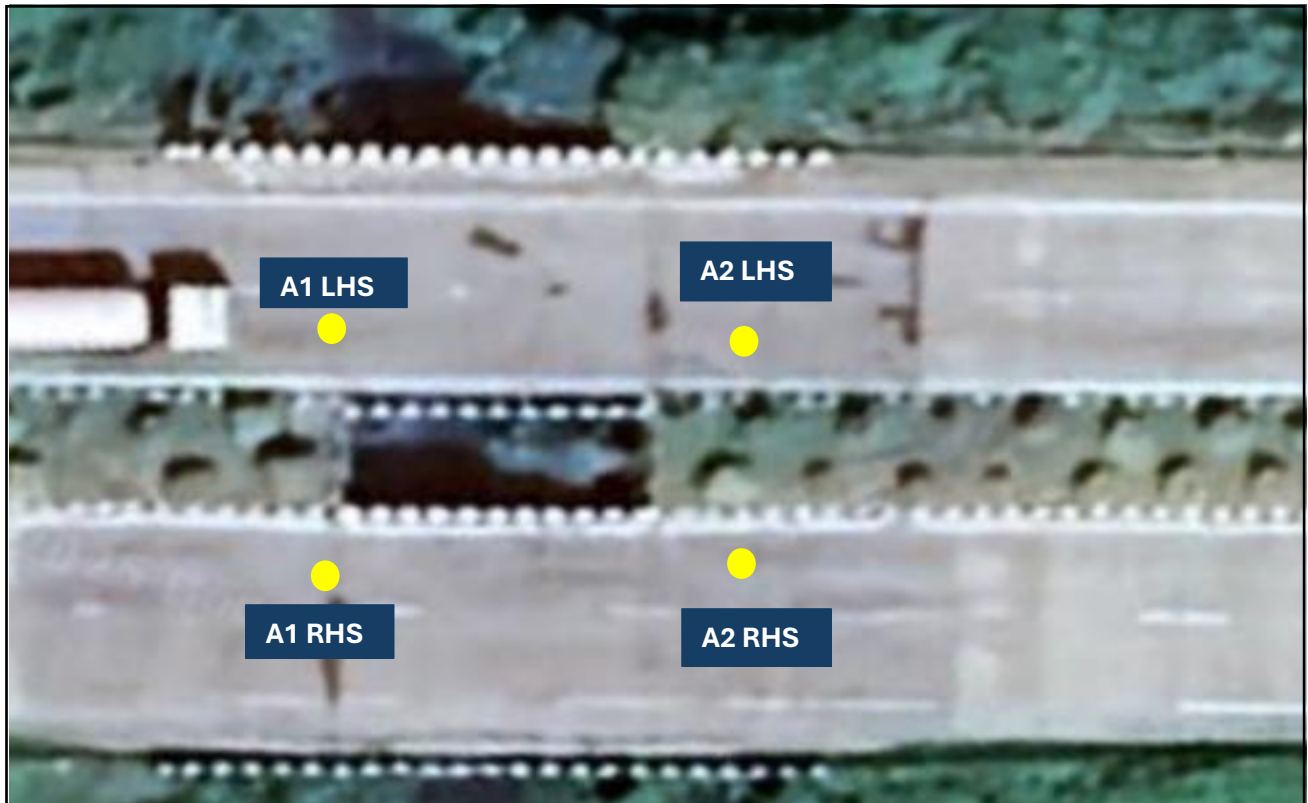


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

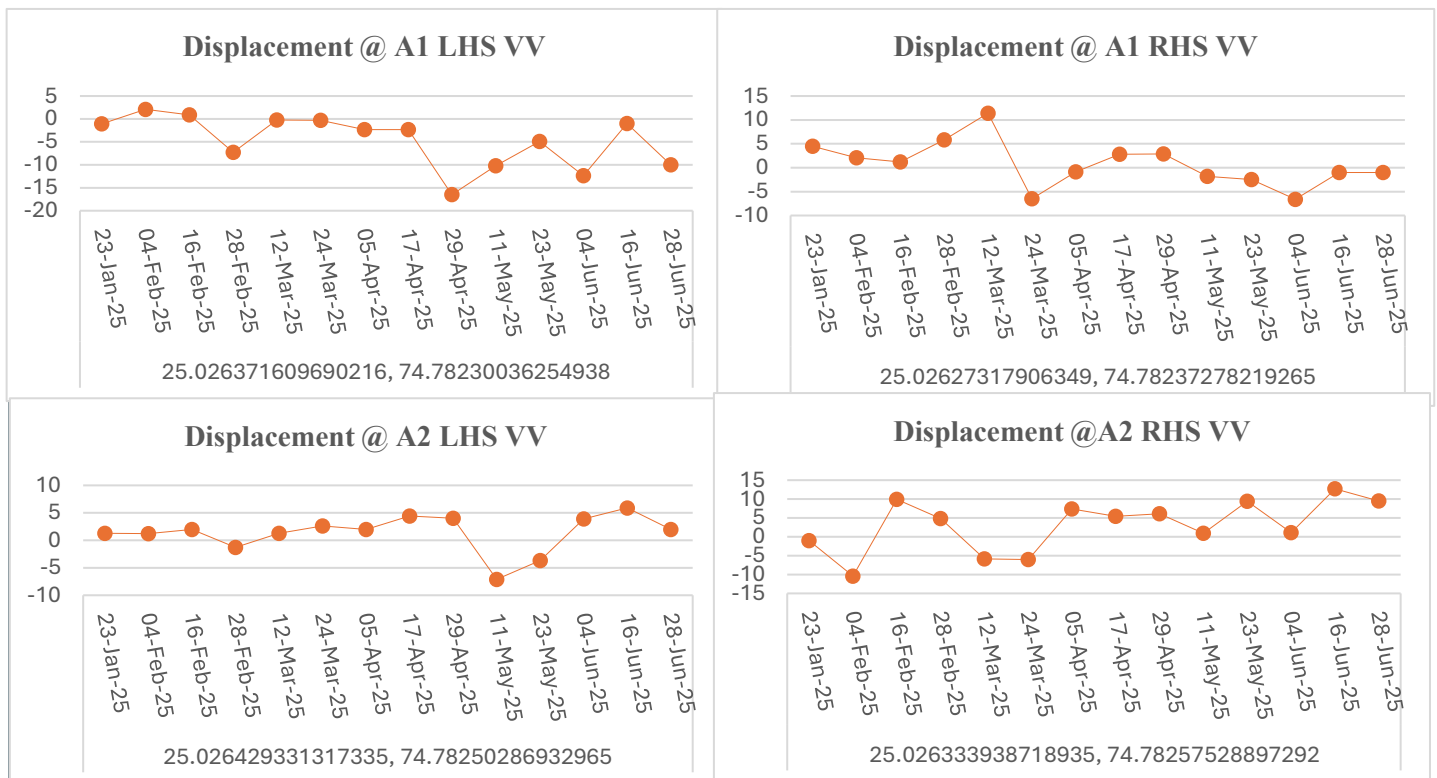
MNB – CH-926+250

Length – 27.85M

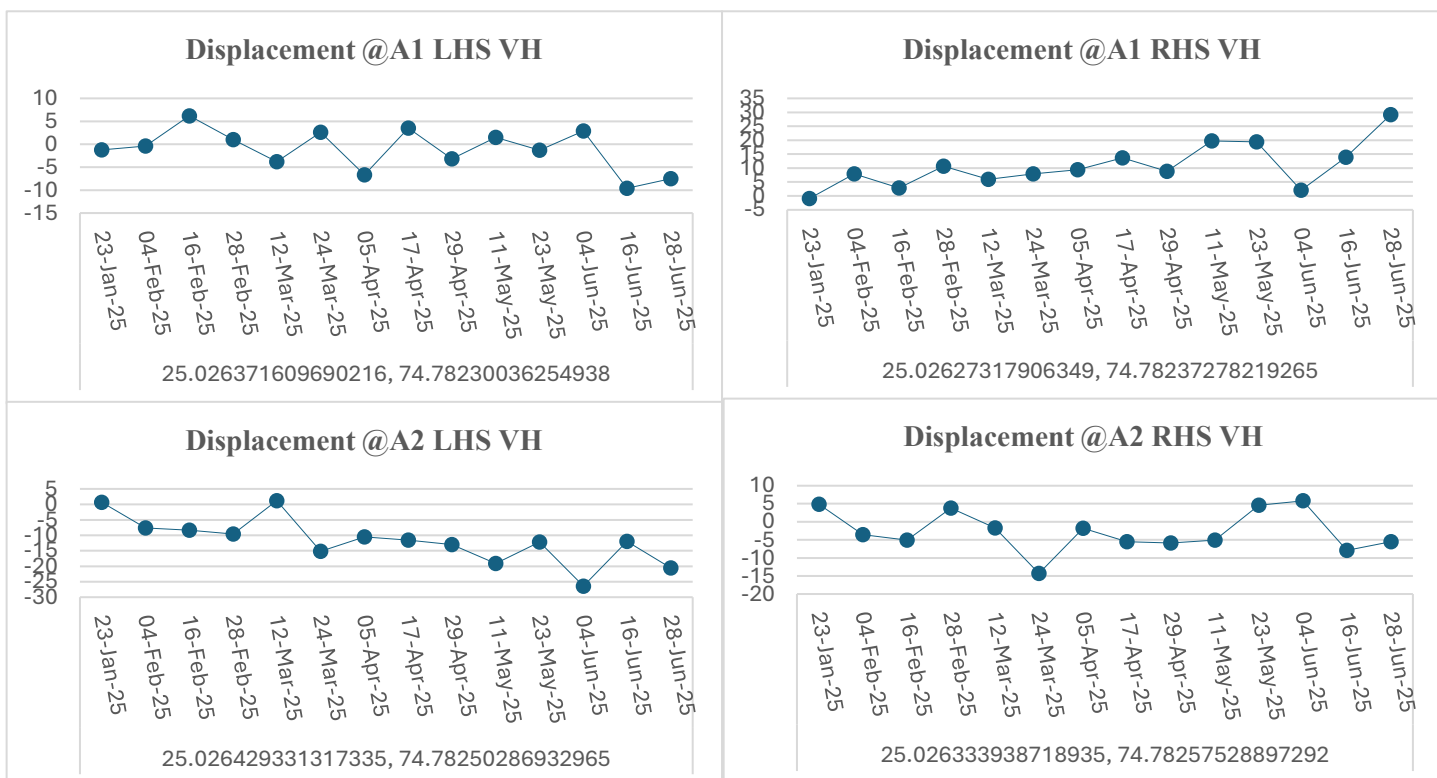


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

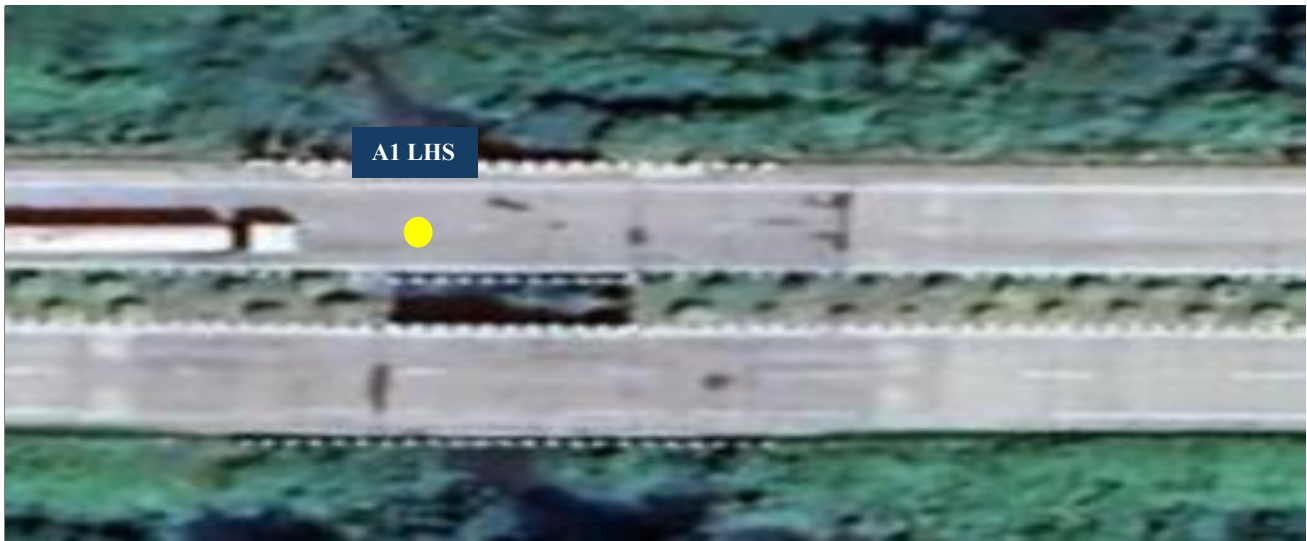
3.1 VV Displacement from 23 Jan 2025 to 28 June 2025



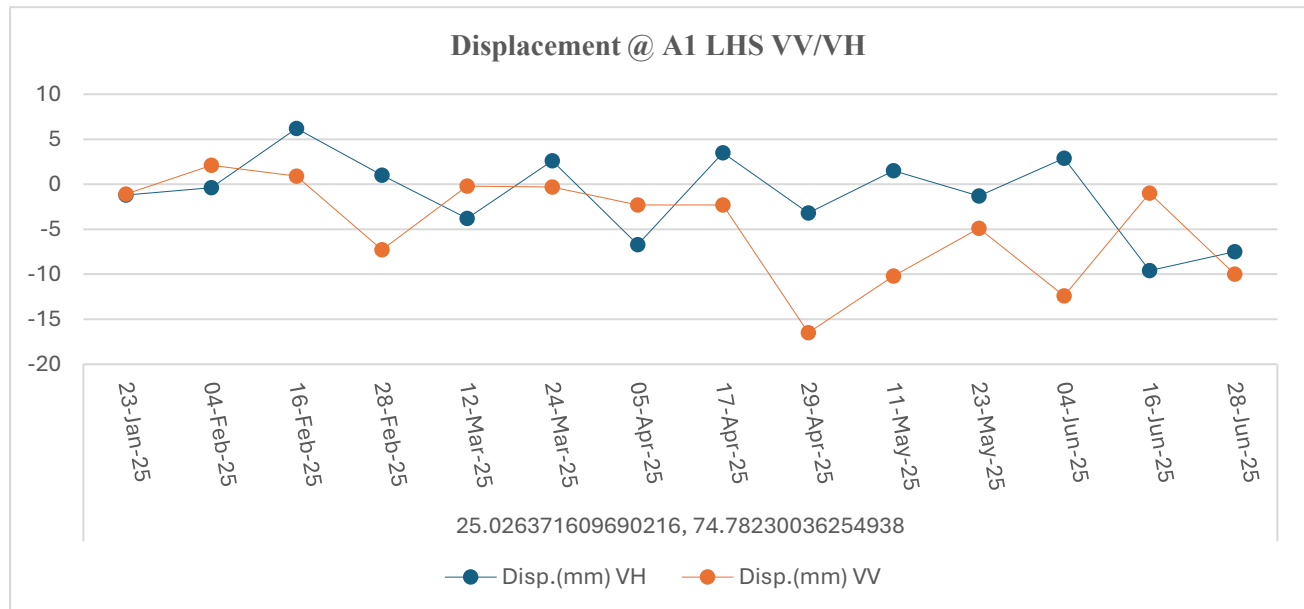
3.2 VH Displacement from 23 Jan 2025 to 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.026371609690216, 74.78230036254938)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** fluctuates between slight uplift and moderate settlement, generally ranging from approximately +2 mm to -17 mm.
- **Horizontal Displacement (VH)** shows alternating north-south movement, varying roughly between +6 mm and -10 mm. While no monotonic trend is evident, the data reflects ongoing lateral flexibility at A1 LHS, with repeated reversals indicating a response to variable loading or environmental influences.

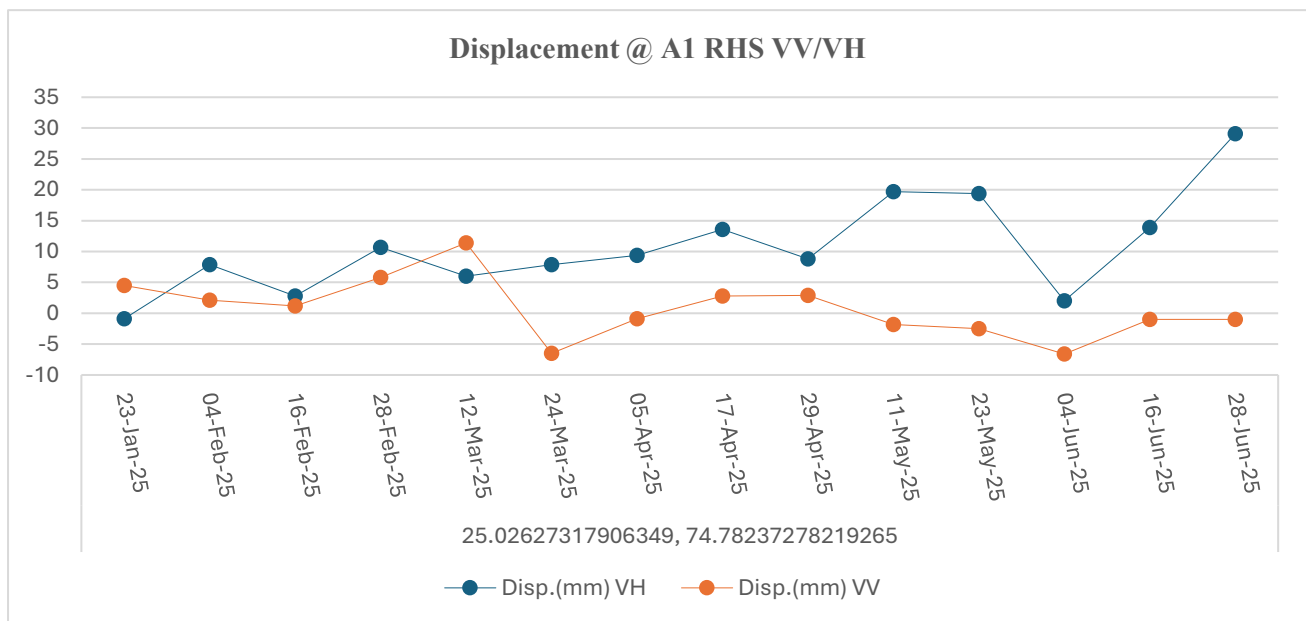
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+6 mm), while VV remains close to neutral, indicating a short-term lateral response without corresponding vertical settlement.
- **March-April 2025:** VV drops sharply to approximately -15 to -17 mm, representing the largest vertical settlement event in the monitoring period, suggesting a transient load or foundation response.
- **June 2025:** Both VV and VH show renewed negative values, though with reduced magnitude, suggesting partial stabilization following earlier displacement extremes.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.02627317906349, 74.78237278219265)



Key Observations:

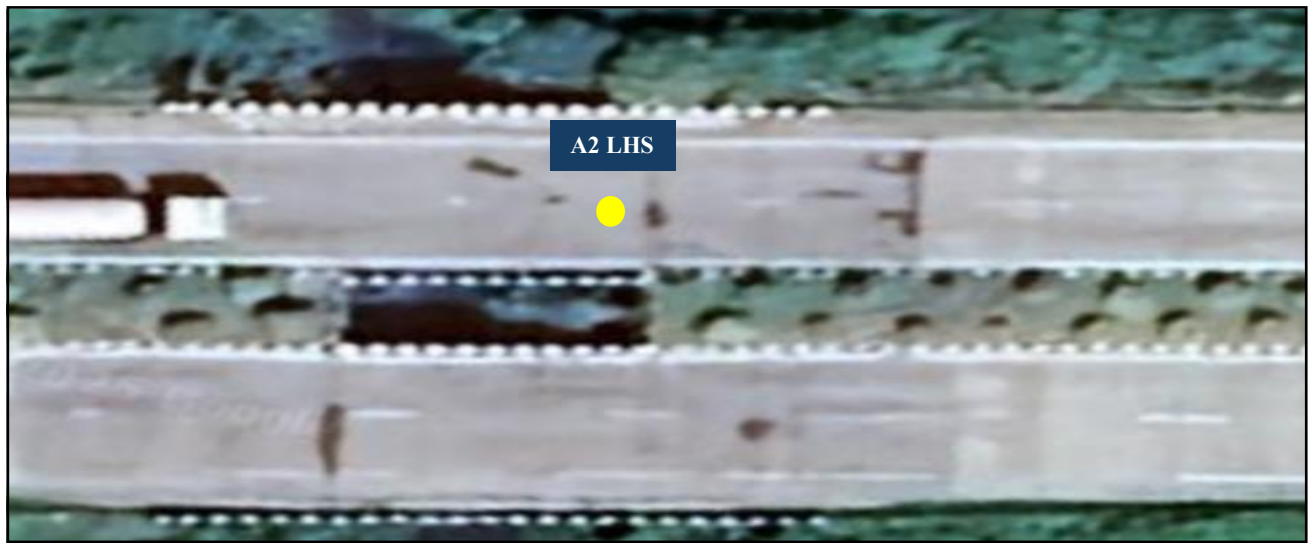
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** shows a clear increasing trend, varying from near 0 mm in January 2025 to approximately +28–30 mm by late June 2025, indicating significant and accumulating lateral movement at A1 RHS.
- **Vertical displacement (VV)** remains relatively low in magnitude throughout the monitoring period, generally ranging between approximately +5 mm and –8 mm

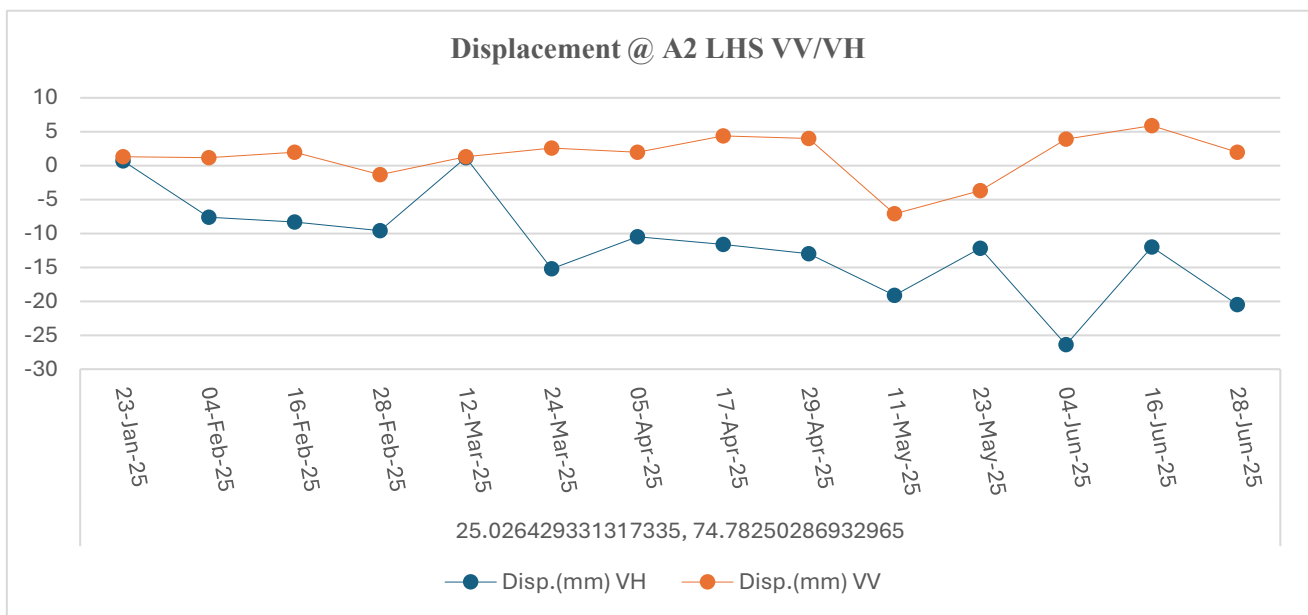
Critical Events & Anomalies:

- **February 2025:** VH increases to around +10–12 mm, while VV remains slightly positive, marking the onset of noticeable lateral displacement.
- **June 2025:** VH reaches its maximum observed value representing the most significant lateral displacement event within the monitoring period, while VV remains modest and stable.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.026429331317335, 74.78250286932965)



Key Observations:

Displacement Magnitude Patterns:

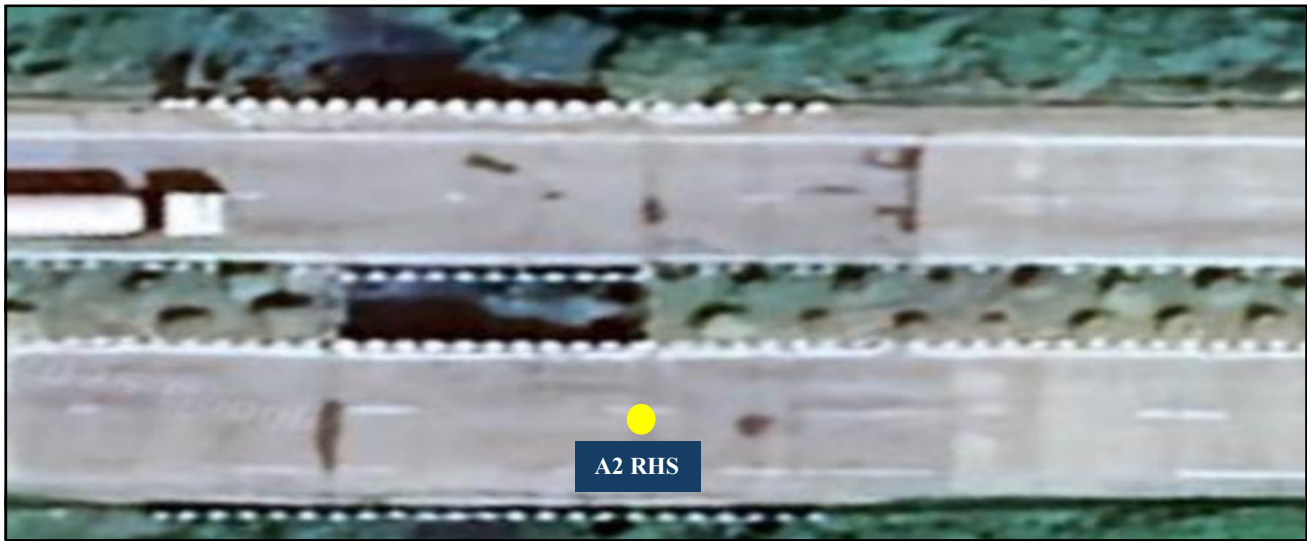
- **Horizontal displacement (VH)** is consistently negative with increasing magnitude, ranging from about –5 mm to –26 mm. The trend shows progressive lateral movement in the negative direction, with several sharp drops indicating episodic lateral shifts.
- **Vertical Displacement (VV)** remains predominantly positive, generally ranging from approximately 0 mm to +6 m. While minor fluctuations are present, the overall behaviour indicates mild vertical uplift with no evidence of progressive settlement at A2 LHS.

Critical Events & Anomalies:

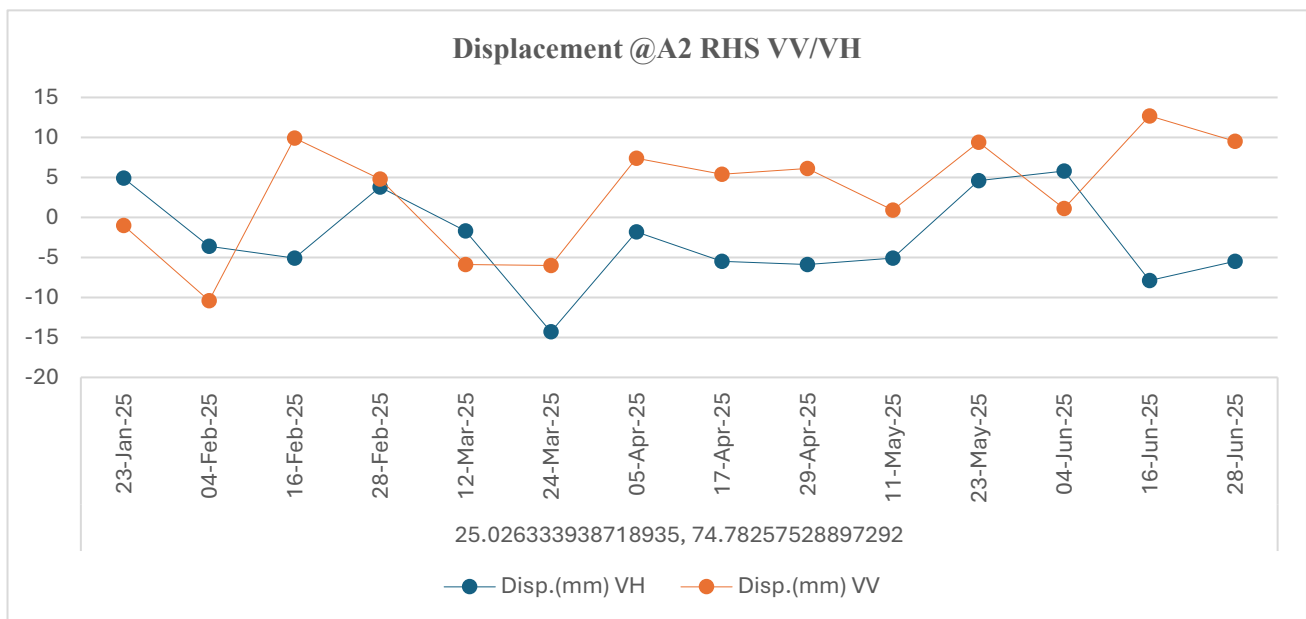
- **Feb 2025:** VH decreases steadily from –7 mm to –10 mm, indicating an early phase of lateral drift. VV remains near zero to slightly positive, suggesting vertical stability during early lateral movement.

June 2025: VH partially recovers to –12 mm before dropping again to –20 mm, indicating continued lateral instability. VV peaks near +6 mm and then reduces slightly remaining predominantly positive through the end of monitoring.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.026333938718935, 74.78257528897292)



Key Observations:

Displacement Magnitude Patterns:

- Vertical (VV) displacement** shows moderate variability with predominantly positive values, ranging approximately from -10 mm to +13 mm.
- Horizontal (VH) displacement** fluctuates across both positive and negative ranges, varying from about +5 mm to -14 mm. The overall behaviour is oscillatory, with no long-term directional drift, reflecting localized lateral response at A2 RHS.

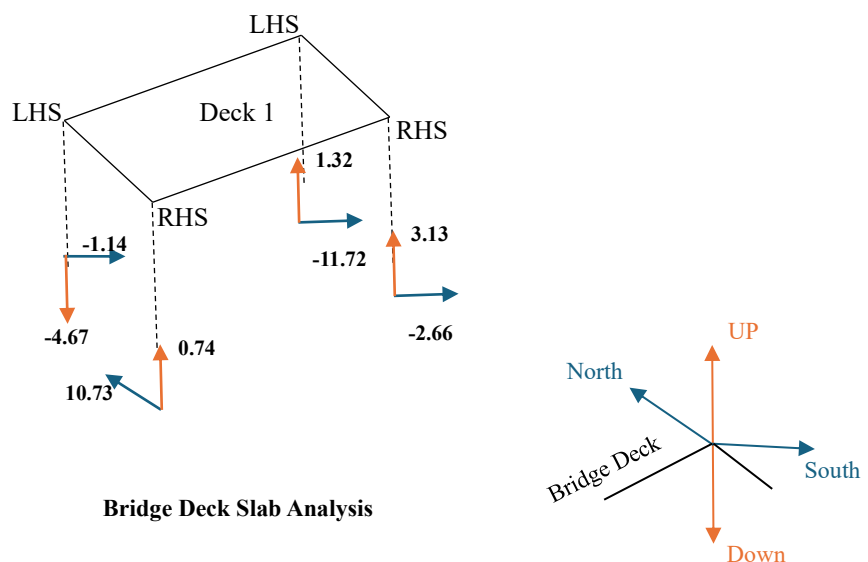
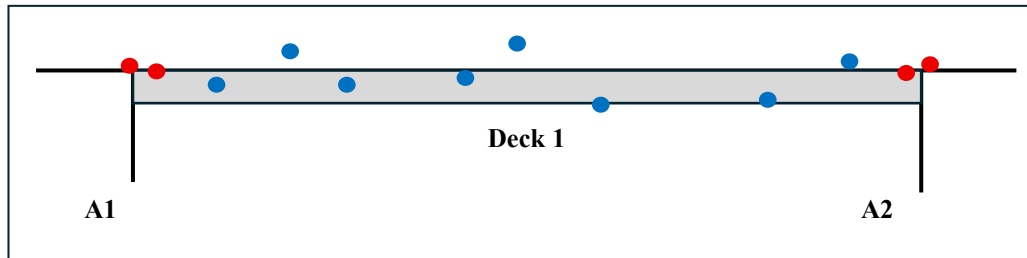
Critical Events & Anomalies:

- February 2025:** VV drops sharply to -10 mm, marking the first significant vertical settlement event. VH is mildly negative (-3 mm), indicating limited lateral involvement during early settlement.
- June 2025:** VH remains negative (-6 mm), while VV stays positive (+9 mm), closing the monitoring period with continued vertical uplift and lateral restraint.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The deck slab shows differential vertical movement across LHS and RHS supports, with vertical displacements ranging approximately from -4.67 mm to $+3.13$ mm. The RHS of the deck exhibits net upward movement, while the LHS shows localized downward displacement, indicating a non-uniform vertical response across the slab width.

Horizontal Displacement Behaviour:

- Horizontal displacements indicate clear directional variation between LHS and RHS. The LHS shows predominant northward movement (10.73 mm) with a minor westward component (-1.14 mm), indicating net translation toward the north-west direction.

CONCLUSION:

- The combined vertical uplift-settlement contrast and opposing horizontal displacement vectors indicate that Deck 1 is undergoing torsional deformation rather than rigid-body translation. The deck slab behaviour reflects rotation about its longitudinal axis, likely driven by uneven load distribution, thermal effects, or differential bearing performance.
- While the displacement magnitudes remain within moderate limits, the presence of torsional distortion warrants continued monitoring to assess whether the observed behaviour is stabilizing or progressive.